MERGED CANONICAL FORM · PHASE B

Section 8 Cosmology

The Open Universe

§8

The ninth deliverable of Phase B: a fully merged, end-to-end rewrite of Section 8 against Construction Spec v1.0. Applies the relational architecture to the universe as a whole. Establishes **Layer placement + FIREWALL** (P1) — cosmology is Layer B→C (expansion = SOE rate, Friedmann = MOE descent on FLRW), with cross-sector gravity explicitly quarantined; seven cosmology Q-items surfaced with explicit dependency tags. Derives **open extension as cosmic expansion**: $H(t) = \gamma_{eff}(t) = (1/V)\cdot dV/dt$ with **Corollary 8.1.2** $H(t) \leq \gamma_{max} = 1/\varepsilon$ (P2 — §2.10 RESOLVED). Derives **Friedmann-like dynamics from MOE descent** on FLRW, with $\Lambda_B = 0$ forcing the no-Λ form (P3 — §5.4.2 RESOLVED). Decomposes the **dark sector** into three parts: relational burden T(rel) as primary dark matter (P4 — §5.1.2 RESOLVED), cross-sector gravity quarantined, dark energy = SOE frontier pressure (NOT cosmological constant). Reconstructs the **early universe**: initial kinematic state ω_{kin} with maximal B_Δ (P5 — §3.6.3 RESOLVED), inflation = rapid SOE extension at γ_{max} + MOE deepening (Q-3), sector formation epochs (§7.4/Q-4 PARTIAL). Restates **Λ_B = 0 EXACT** as Theorem 8.5.1 (P6 — Q-9 PARTIAL). Closes with **ΛCDM/Lorentz IR/Q-4 status tables + nine consolidated guardrails** (P7).

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PHASE B · Section 8 Merge

SCOPE 7 Subsections · Layers B→C, Λ quarantined · 15 Units

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LAYER PLACEMENT + FIREWALL (P1) | COSMIC EXPANSION = SOE RATE (P2) | §2.10 RESOLVED | FRIEDMANN = MOE DESCENT (P3) | §5.4.2 RESOLVED | DARK SECTOR THREE-PART (P4) | §5.1.2 RESOLVED | EARLY UNIVERSE (P5) | §3.6.3 RESOLVED | Λ_B = 0 EXACT (P6) | NINE GUARDRAILS (P7)

Preamble — How to Read This Section

This document is the merged canonical form of Section 8 of the Reconciliation Causal Framework (RCF). It is the ninth deliverable of Phase B as specified in *RCF Unified Construction Specification v1.0*, and it builds directly on the closed foundations of Sections 0–7. Section 0 produced the kinematic algebra $\mathscr{A}_{kin}$, the GNS representation (π_ω, $\mathscr{H}_\omega$, Ω_ω), the Reconciliation Propagator $\mathbf{R}_t$ = SOE ∘ MOE, and the physical sub-algebra $\mathscr{A}_{phy}$. Section 1 introduced the strict partial order of causal dependency ≺ (§1.1.3), the two-scale (SOE/MOE) speed limit $c = \gamma \cdot \ell_0$ (§1.3), and open extension as the causal frontier (§1.4). Section 2 constructed the correlation kernel K_ω, the cubic volume element $\mathscr{V}_\omega$(A, B, C) (Def 2.10), and the D=3 closure (Thm 2.3.3). Section 3 derived the constraint burden $B[\rho] = Tr(\rho\ \mathbb{F})$ (§3.1) and the burden-clock (§3.2, §3.3). Section 4 reconstructed the matter layer with mass-burden identity $m \equiv B_0$ (§4.2.8). Section 5 derived the gravitational layer: burden tensor $\Theta_{\mu\nu}^{(B)}$ sources curvature, $\Lambda_B = 0$ EXACT (Thm 5.5), κ_B derived from saturation limit, and the three-channel burden decomposition (mode/interaction/relational) at §5.1.2. Section 6 examined black holes as unreconciled relational sectors with Bekenstein-Hawking entropy and Hawking-like emission. Section 7 closed the probability layer: record sectors, stable record separation, redundant m-robust classicality, normalized record weights, sectorwise zero-decomposition, Z-envariance as MOE fixed-point symmetry, and the Born rule $p_i = |c_i|^2$ DERIVED (Thm 7.5, T-4 STRENGTHENED), protected by the FIREWALL guardrail.

Section 8 now applies the relational architecture to the universe as a whole. The framework has constructed spacetime, matter, gravity, black holes, and probability entirely from relational constraint dynamics — but every application so far has been local: a patch of spacetime, a particle, a star, a black hole, a measurement event. Cosmology asks the global question: **what is the universe, taken as a whole, in the reconciliation picture?** The architectural answer is decisive. Under the SOE/MOE decomposition, the universe expands because new open extensions continuously incorporate new events at the causal frontier (Layer B→C, expansion = SOE rate), and its large-scale dynamics are governed by MOE descent on the emergent cosmological metric (Layer C, Friedmann = Euler-Lagrange of MOE on FLRW). The two scales separate cleanly. The dark sector has two distinct mechanisms — relational burden (derived, $T^{(rel)} = [<\hat{C}_\alpha, \Pi_{net}]$) and cross-sector gravity (quarantined conjecture) — and dark energy is reinterpreted as the dynamic pressure of open extension at the causal frontier (NOT a cosmological constant, since $\Lambda_B = 0$ is exact by Thm 5.5).

The structure follows the spec's source map (Table 4.1, rows 8.3 and 8.4 — both QUARANTINE per Spec Ch. 12.2) and the Gen 1 master manuscript RCF_n.txt §8.0–8.5, augmented throughout by Section_8_Cosmology_2.txt for the SOE/MOE Layer placement (P1), the Friedmann-as-MOE-Euler-Lagrange interpretation (P3), the three-part dark-sector decomposition (P4), and the architectural-position table mapping each structure to its layer. Per Spec Ch. 13.3, **quarantine items (Table 12.2) appear only in Section 8 with their Q-IDs**; seven cosmology-relevant items (Q-1, Q-2, Q-3, Q-4, Q-9, Q-13, Q-17) are surfaced here with explicit dependency tags. Each subsection opens with a layer badge identifying its position in the L→Q→C→Q′ emergence ladder (Section 8 occupies Layers B→C, with Layer A→C cross-sector gravity explicitly quarantined). Body text is ported verbatim where possible; rewritten passages are flagged inline with a spec chapter reference.

Dependency contract with Sections 0–7

Section 8 depends on seven structures from the closed foundation: **(i)** the Reconciliation Propagator $\mathbf{R}_t$ = SOE ∘ MOE and the SOE/MOE decomposition from §0.4 — cosmic expansion is the SOE frontier rate (P2), Friedmann dynamics are MOE descent on FLRW (P3); **(ii)** open extension as the causal frontier from §1.4 — the universe expands because new open extensions continuously incorporate new events; **(iii)** the two-scale speed limit $c = \gamma \cdot \ell_0$ and $\gamma_{max} = 1/\varepsilon$ from §1.3 — Corollary 8.1.2 caps $H(t) \le \gamma_{max}$; **(iv)** the cubic volume element $\mathcal{V}_\omega(A, B, C)$ from §2.10 and the D=3 closure (Thm 2.3.3) — $V(t) \propto a(t)^3$ is the SOE-invariant volume growth; **(v)** the three-channel burden tensor decomposition $\Theta_{\mu\nu}^{(B)} = T^{(mode)} + T^{(int)} + T^{(rel)}$ from §5.1.2 — relational burden $T^{(rel)}$ is the derived dark-matter mechanism (P4); **(vi)** $\Lambda_B = 0$ EXACT (Thm 5.5) and $\kappa_B = C/(\Pi_{max} \cdot \ell_0^2)$ from §5.4.2 / §5.5 — forces Friedmann form WITHOUT cosmological constant term (P3, P6); **(vii)** stable record separation (Thm 7.1) and sector weights $p_k = \mathrm{Tr}(P_k\, \rho_{kin})$ from §7.1 / §7.4 — sector formation epochs (§8.4.3) and the matter-antimatter asymmetry Q-4 mechanism depend on these. All seven dependencies are one-way: Section 8 does not modify any structure of Sections 0–7.

Forward-reference contracts RESOLVED here

Section 8 closes four forward references left open by Sections 2, 3, 5, and 7: **(a)** §2.10 cubic correlation volume $\mathcal{V}_\omega$ ↔ cosmological scale factor a(t) — resolved in §8.1 (P2); **(b)** §3.6.3 cosmological initial condition → §8 — resolved in §8.4.1 (P5, Initial Kinematic State); **(c)** §5.1.2 dark matter = relational burden → §8 — resolved in §8.3.1 (P4, primary mechanism); **(d)** §5.4.2 $\Lambda_B = 0$ + dark energy reinterpretation → §8 — resolved in §8.2 + §8.3.3 + §8.5 (P3, P4, P6). One forward reference is PARTIALLY resolved: **(e)** §7.4 sector weight asymmetry p_k → §8 (Q-4 matter-antimatter asymmetry) — sector formation epochs (§8.4.3) supply the structural mechanism, but formal derivation of the asymmetry remains open (Q-4 partial → §9 audit).

FIREWALL guardrail — REAFFIRMED for cosmology

The FIREWALL guardrail (§5.0 P1, §6.0 P1, §7.0 P1) is reaffirmed for cosmology: **probability belongs to Layer A** (branch weights $p_k = \mathrm{Tr}(P_k\, \rho_{kin})$), **gravity belongs to Layer C** (MOE hydrodynamics sourced by $\mathrm{Tr}(\rho\, F)$), and **burden linearity (§0.3, Property 3)** guarantees $\mathrm{Tr}(\rho_{kin}\, F) = \Sigma_k\, p_k\, \mathrm{Tr}(\rho_k\, F)$ — a PROVEN IDENTITY of the linear functional, NOT an averaging-over-outcomes applied to source cosmic curvature. The notation $\Sigma_k\, p_k\, T^{(k)}$ in a cosmological Friedmann source is just regrouping terms of a single linear functional; it does NOT mean "averaging sectors by Born-rule weights to source gravity." Cross-sector gravity (§8.3.2) is explicitly quarantined precisely because it would require crossing the FIREWALL. No conflation of probability and gravity is permitted at any cosmological scale.

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§8.0 Purpose and Layer Placement

Layer B→C · P1

Source: Section_8_Cosmology_2.txt §8.0 + RCF Unified Construction Specification v1.0 Ch. 12.2 (17-point quarantine list).

The framework's final application is cosmology — the large-scale behavior of SOE extension and MOE descent applied to the universe as a whole. The universe expands as new open extensions are incorporated at the causal frontier; the dark sector has two distinct mechanisms, one derived (relational burden as dark matter) and one quarantined (cross-sector gravity); and dark energy is reinterpreted as the dynamic pressure of open extension, NOT as a cosmological constant. Under the SOE/MOE decomposition, cosmic expansion is the SOE incorporation rate at the causal frontier (Layer B→C), and the Friedmann dynamics are the MOE descent equations on the emergent cosmological metric (Layer C). The two scales separate cleanly.

Section 8 is the largest-scale application of the SOE/MOE decomposition and the canonical home of the framework's quarantined phenomenology. Per Spec Ch. 13.3, **quarantine items (Table 12.2) appear only in Section 8 with their Q-IDs**. Seven cosmology-relevant items are surfaced here: Q-1 (dark matter), Q-2 (dark energy), Q-3 (inflation), Q-4 (matter-antimatter asymmetry), Q-9 (cosmological constant problem), Q-13 (ΛCDM recovery), and Q-17 (Lorentz invariance in deep IR). Each is labeled with its dependency contract; closed items are marked CLOSED, partially-closed items are marked PARTIAL, and open items defer to §9 audit.

#	Quarantined Claim	Layer	Dependency	Status
Q-1	Dark matter = relational burden	B→C (derived)	T-7 (Poisson recovery) + numerical sim of MOE descent on FLRW	OPEN — structural match only
Q-2	Dark energy = SOE frontier pressure	C (conjectural)	Λ_B=0 (CLOSED §5.5) + FLRW closure (PARTIAL §8.2)	PARTIAL — Λ closed, FLRW partial
Q-3	Inflation = early-SOE rapid expansion	A→C (conjectural)	Cosmological initial conditions; separate from core framework	OPEN — structural mechanism only
Q-4	Matter-antimatter asymmetry	A→C	T-4 (Born rule, CLOSED §7) + sector formation (§8.4.3)	PARTIAL — T-4 closed, mechanism proposed
Q-9	Cosmological constant problem	C	Λ_B=0 derivation (CLOSED §5.5); T-2 stable-mode	PARTIAL — Λ closed, T-2 pending
Q-13	Recovery of standard cosmology (ΛCDM)	B→C	FLRW closure (§8.2) + dark sector items 1–2 (Q-1, Q-2)	PARTIAL — FLRW closed, dark sector open
Q-17	Lorentz invariance in deep IR	B→C	T-1 (γ derivation, §1.3) + Lorentzian signature (Thm 5.3)	PARTIAL — T-1 ✓, signature ✓, formal SR equivalence pending

Table 8.0.1 — Seven cosmology-relevant quarantine items with dependency status. Per Spec Ch. 12.2, all 17 Q-items appear only in Section 8 with their Q-IDs; the remaining 10 (Q-5, Q-6, Q-7, Q-8, Q-10, Q-11, Q-12, Q-14, Q-15, Q-16) are addressed in their respective subject sections (matter, quantum gravity, black holes, probability) and surfaced for final audit in §9.

P1 — Layer placement + FIREWALL

Cosmology occupies **Layer B→C** (cosmic expansion = SOE frontier rate, Friedmann dynamics = MOE descent on FLRW), with **Layer A→C cross-sector gravity explicitly quarantined** (§8.3.2). The FIREWALL guardrail is REAFFIRMED: **probability belongs to Layer A** (branch weights p_k = $\mathrm{Tr}(P_k\ \rho_{kin})$), **gravity belongs to Layer C** (MOE hydrodynamics sourced by $\mathrm{Tr}(\rho\ F)$), and burden linearity (§0.3, Property 3) guarantees $\mathrm{Tr}(\rho_{kin}\ F) = \Sigma_k\ p_k\ \mathrm{Tr}(\rho_k\ F)$ — a PROVEN IDENTITY of the linear functional, NOT an averaging-over-outcomes applied to source cosmic curvature. Consistent with §5.0 P1, §6.0 P1, and §7.0 P1: no conflation of probability and gravity is permitted at any cosmological scale. The cross-sector gravity conjecture (§8.3.2) is quarantined precisely because it would require crossing the FIREWALL.

The architectural position table at §8.5 maps each cosmological structure to its layer. The conceptual chain — open extension → cosmic expansion → cubic volume growth → Friedmann-like dynamics → dark sector (relational burden + quarantined cross-sector + dark energy) → early universe (kinematic state + inflation + sector formation) → $\Lambda_B = 0$ → ΛCDM/Lorentz IR/Q-4 status — is acyclic, with every claim traced either to a closed theorem in Sections 0–7 or to an explicitly labeled quarantine item.

§8.1 Open Extension as Cosmic Expansion (SOE Rate)

Layer B→C · P2

Source: Section_8_Cosmology_2.txt §8.1 (Conjecture 8.1.1, Corollary 8.1.2). Resolves forward reference from §2.10.

The expansion of the universe is the continuous incorporation of new events at the causal horizon of our reconciliation sector. In RCF terms, this is open extension (§1.4) operating at the cosmic frontier: new open extensions continuously incorporate new events at speed $c = \gamma \cdot \ell_0$ (§1.3), growing the reconciled volume. The Hubble parameter is the large-scale reconciliation rate — the coarse-grained SOE extension rate at the causal frontier. This identification is structural, not phenomenological: it follows from the definition of the Reconciliation Propagator $\mathbf{R}_t$ = SOE ∘ MOE (§0.4) applied at cosmological scale, where the SOE step incorporates new open extensions and the MOE step reconciles them into the physical sector.

Conjecture 8.1.1 (Expansion = SOE Frontier Rate).

The Hubble parameter H(t) is the coarse-grained SOE extension rate at the causal frontier:

$$H(t) = \gamma_{eff}(t) = (1/V) \cdot dV/dt \tag{8.1}$$

where V(t) is the reconciled volume — the number of SOE-extended reconciliation cells within the causal horizon. The universe expands because new open extensions continuously incorporate new events at speed $c = \gamma \cdot \ell_0$. This identification closes the forward reference from §2.10: the cubic correlation volume $\mathscr{V}_\omega(A, B, C)$ constructed in Section 2 (Definition 2.10) is the local precursor of the cosmological scale factor a(t); at cosmological scale, the reconciled volume V(t) is the integral of $\mathscr{V}_\omega$ over the causal horizon, and the SOE extension rate is the Hubble flow.

Corollary 8.1.2 (Expansion Rate Limit).

The SOE flux capacity is bounded above by $\gamma_{max} = 1/\varepsilon$ (§1.3), where ε is the SOE time-resolution floor. Coarse-graining to cosmological scale preserves the bound:

$$H(t) \le \gamma_{max} = 1/\varepsilon \tag{8.2}$$

The universe cannot extend faster than one reconciliation cell per SOE time-step. This places a hard upper bound on the Hubble parameter at every epoch, including the inflationary epoch — inflation cannot exceed γ_{max}, and the apparent super-luminal expansion of distant galaxies is consistent with this bound because it reflects growing separation between reconciliation cells, not motion through a pre-existing spacetime.

Forward reference from §2.10 — RESOLVED

Section 2 v1.0 left a forward reference from §2.10 (Definition 2.10, cubic correlation volume $\mathcal{V}_\omega$) to the cosmological scale factor a(t). **This merged Section 8 resolves that forward reference**: the cubic correlation volume is the local precursor of a(t). At cosmological scale, $V(t) = \int_{\text{causal horizon}} \mathcal{V}_\omega \, d^3x$ is the reconciled volume, and the SOE extension rate $\gamma_{eff}(t) = (1/V)\cdot dV/dt$ is the Hubble parameter. The D=3 closure (Thm 2.3.3) protects $V(t) \propto a(t)^3$ as an SOE invariant (Conjecture 8.2.1, P3). No new primitives are introduced; the cosmological scale factor is the large-scale coarse-graining of the cubic correlation volume.

Q-17 (Lorentz invariance in deep IR) — PARTIAL

Quarantine item Q-17 (recovery of Lorentz invariance in the deep IR limit) is partially addressed by §8.1: **(i)** the two-scale speed limit $c = \gamma \cdot \ell_0$ (§1.3, T-1) establishes the universal velocity ceiling that becomes the speed of light in the deep IR; **(ii)** the emergent Lorentzian signature (Theorem 5.3, derived from positive spatial correlation + negative burden lapse) ensures the metric takes the (−,+,+,+) form required by special relativity. **What remains open**: a formal proof that the deep-IR limit of emergent correlation geometry is exactly Minkowski spacetime with full Lorentz symmetry (not just metric signature). This is a theorem-target, not a structural obstacle; the framework's primitives (γ, ℓ_0, K_ω) all support the identification. Q-17 status: PARTIAL → §9 audit.

§8.2 Friedmann-like Dynamics from MOE Descent

Layer C · P3

Source: Section_8_Cosmology_2.txt §8.2 (Conjecture 8.2.1, Conjecture 8.2.2). Resolves forward reference from §5.4.2.

With the Hubble parameter identified as the SOE extension rate (§8.1), the next question is the dynamical equation governing a(t). The RCF answer is structural: the Friedmann-like equation is the Euler-Lagrange equation of MOE descent on the emergent cosmological metric. Just as the Einstein equation $G_{\mu\nu} = \kappa_B \, \Theta^{(B)}_{\mu\nu}$

emerged as the Euler-Lagrange equation of MOE descent on the space of emergent metrics in §5.3 (Theorem 5.4), so the Friedmann equation emerges as the Euler-Lagrange equation of MOE descent on the FLRW family of cosmological metrics. The burden tensor $\Theta_{\mu\nu}^{(B)}$ with three-channel decomposition (mode/interaction/relational, §5.1.2) sources the cosmological dynamics; $\Lambda_B = 0$ (Thm 5.5) forces the Friedmann form WITHOUT a cosmological constant term.

Conjecture 8.2.1 (Cubic Volume Growth).

Since spatial geometry is 3D (Theorem 2.3.3) and protected as a SOE invariant, the reconciled volume grows as:

$$V(t) \propto a(t)^3 \tag{8.3}$$

where a(t) is the cosmological scale factor. The D=3 closure is an SOE invariant — the spatial dimensionality is locked by the cubic correlation volume construction (§2.10) and preserved under SOE extension at the causal frontier. This rules out higher-dimensional cosmological embeddings and identifies the observed 3+1 spacetime as the framework's structural prediction, not an input.

Conjecture 8.2.2 (Friedmann-like Equation from MOE Descent).

MOE gradient descent on the cosmological metric yields:

$$(\dot{a}/a)^2 = (8\pi G/3) \cdot \rho_B - k/a^2 \tag{8.4}$$

where $\rho_B = \Sigma_{\text{channels}}\, \rho_B^{(\text{channel})}$ includes all three burden channels (mode, interaction, relational), and k is the spatial curvature from the sector's geometry. This is the Euler-Lagrange equation of MOE descent on the FLRW metric. Note the absence of a Λ term: $\Lambda_B = 0$ EXACT (Theorem 5.5) forces the Friedmann form WITHOUT cosmological constant. The observed cosmic acceleration is NOT a Λ contribution; it is the dynamic residual burden pressure of open extension (Conjecture 8.3.3).

> **Forward reference from §5.4.2 — RESOLVED**
>
> Section 5 v1.0 left a forward reference from §5.4.2 ($\Lambda_B = 0$ EXACT, Theorem 5.5, plus the reinterpretation of dark energy as the dynamic pressure of Open Extension). **This merged Section 8 resolves that forward reference**: in §8.2 (P3), $\Lambda_B = 0$ forces the Friedmann form WITHOUT cosmological constant term; in §8.3.3 (P4), dark energy is reinterpreted as the dynamic pressure of open extension at the causal frontier (NOT a cosmological constant); in §8.5 (P6), Theorem 8.5.1 restates $\Lambda_B = 0$ EXACT in the cosmological context. The 120-order-of-magnitude cosmological constant problem is structurally resolved: there is no Λ to fine-tune, because the Master-Zero condition $\omega(\mathcal{M}) = 0$ is an EXACT asymptotic constraint achieved by MOE descent (§5.5, Theorem 5.5), not a tuned parameter.

> **Q-13 (ΛCDM recovery) — PARTIAL**

> Quarantine item Q-13 (recovery of standard cosmology, ΛCDM) is partially addressed by §8.2: **(i)** FLRW closure is STRUCTURALLY CLOSED — the Friedmann-like equation (8.4) is the Euler-Lagrange of MOE descent on FLRW, with the burden tensor replacing the standard stress-energy; **(ii)** the $\Lambda = 0$ prediction is structurally forced (Thm 5.5), differing from ΛCDM's small-but-nonzero Λ — the framework predicts that what ΛCDM calls "dark energy" is in fact dynamic residual pressure (§8.3.3), recoverable as an effective Λ only in the coarse-grained, time-averaged limit. **What remains open**: quantitative recovery of ΛCDM amplitudes (H_0, Ω_m, Ω_Λ^{eff}) requires numerical simulation of MOE descent on FLRW, plus closure of Q-1 (dark matter amplitudes) and Q-2 (dark energy effective equation of state). Q-13 status: PARTIAL — FLRW closed, dark sector amplitudes open → §9 audit.

§8.3 Dark Sector — Three-Part Decomposition

Layers B→C, A quarantined · P4

Source: Section_8_Cosmology_1.txt §8.3 + Section_8_Cosmology_2.txt §8.3 (three-part split). Resolves forward reference from §5.1.2.

The dark sector — the discrepancy between observed cosmic dynamics and the visible-matter stress-energy — has two distinct candidate mechanisms in RCF, plus a third for dark energy. The architectural discipline is strict: **relational burden is the primary, derived dark-matter mechanism**; **cross-sector gravity is a quarantined secondary candidate**, NOT a derivation; and **dark energy is the dynamic pressure of open extension**, NOT a cosmological constant. The three are structurally distinct and must not be conflated.

§8.3.1 Primary Mechanism: Relational Burden as Dark Matter

Layer B→C (derived)

Source: Section_8_Cosmology_1.txt §8.3.1 + Section_8_Cosmology_2.txt §8.3.1. Resolves forward reference from §5.1.2.

The relational burden channel $T^{(rel)}$ (Section 5.1.2) is the framework's derived dark-matter mechanism. Generated by the commutator $[\hat{C}_\alpha, \Pi_{net}]$ — the cost of maintaining the background correlation web — it is the MOE residual of the cross-extension correlation network. As a structural consequence of the commutator definition, $T^{(rel)}$ has four properties that match dark-matter phenomenology, derived from RCF primitives rather than imported from observation:

#	Property	Mechanism (derived from $[\hat{C}_\alpha, \Pi_{net}]$)	Phenomenological Match
1	Correlation with matter clustering	Π_{net} is dense where particles (SOE modes) cluster — the commutator burden pools where the network is dense	Dark matter halos correlate with visible matter distribution
2	Non-luminosity	Π_{net} belongs to cross-extension topology, not localized SOE modes — no electromagnetic coupling	Dark matter does not emit, absorb, or scatter light

#	Property	Mechanism (derived from $[\hat{C}_\alpha, \Pi_{net}]$)	Phenomenological Match
3	Halo extension	The correlation kernel K_ω decays smoothly, so $\mathrm{supp}(\rho_{rel}) \supset \mathrm{supp}(\rho_{mode})$ — relational burden extends beyond visible matter	Dark matter halos extend beyond the visible galaxy
4	Gravitational response	$T^{(rel)}$ enters $T_{\mu\nu}^{(B)}$ and sources curvature via $G_{\mu\nu} = \kappa_B T_{\mu\nu}^{(B)}$ (Thm 5.4)	Dark matter gravitates — inferred from rotation curves, lensing, cluster dynamics

Table 8.3.1 — Four structurally derived properties of relational burden $T^{(rel)}$ matching dark-matter phenomenology. These are consequences of the commutator $[\hat{C}_\alpha, \Pi_{net}]$, not imports from dark-matter observation.

These four properties are **consequences of the commutator definition**, not imports from dark-matter phenomenology. The dark-matter-like behavior is a downstream observation, not a design target. This closes the forward reference from §5.1.2 (dark matter = relational burden channel): the relational burden $T^{(rel)}$ is the derived dark-matter mechanism, surfaced as a structural prediction of the three-channel burden decomposition rather than as a tuned addition to the stress-energy tensor.

Forward reference from §5.1.2 — RESOLVED

Section 5 v1.0 left a forward reference from §5.1.2 (three-channel burden decomposition, with relational burden $T^{(rel)}$ identified as the dark-matter mechanism). **This merged Section 8 resolves that forward reference**: in §8.3.1 (P4), the four properties of $T^{(rel)}$ are derived from the commutator $[\hat{C}_\alpha, \Pi_{net}]$ and matched against dark-matter phenomenology. The mechanism is structural — relational burden is the MOE residual of the cross-extension correlation network, sourced by the cost of maintaining constraint compatibility across the network. No new primitives are introduced; dark matter is a prediction of the three-channel burden decomposition.

Q-1 (Dark matter = relational burden) — OPEN

Quarantine item Q-1 is structurally addressed in §8.3.1 but remains OPEN: **(i)** the four derived properties match dark-matter phenomenology qualitatively, but quantitative recovery (rotation curve shapes, halo mass profiles, bullet cluster dynamics) requires numerical simulation of MOE descent on FLRW with realistic correlation kernel K_ω; **(ii)** the Poisson-recovery theorem (T-7) — that the Newtonian limit of relational burden reproduces $\nabla^2\Phi = 4\pi G \cdot \rho_{rel}$ with the correct coupling — is a theorem-target, not yet a closed theorem. Q-1 status: OPEN — structural mechanism closed, quantitative recovery pending T-7 + numerical sim → §9 audit.

§8.3.2 Secondary Candidate: Cross-Sector Gravitational Contribution

Layer A (speculative) — QUARANTINED

Source: Section_8_Cosmology_1.txt §8.3.2 + Section_8_Cosmology_2.txt §8.3.2. Conjecture 8.3.2 — QUARANTINED.

If multiple reconciliation sectors exist beyond ker(M̃), they may contribute gravitationally through cross-sector MOE burden coupling. This would amount to a Layer A → Layer C gravitational leakage between quantum-causally disconnected sectors — precisely the kind of cross-layer mechanism that the FIREWALL guardrail prohibits. The conjecture is therefore explicitly quarantined:

Conjecture 8.3.2 (Cross-Sector Contribution).

If multiple reconciliation sectors exist and interact gravitationally through their burden tensors (Conjecture 1.6.1), then the total source includes a cross-sector term $T_{\mu\nu}^{total} = \oplus_k T_{\mu\nu}^{(k)}$. However, this conjecture is explicitly quarantined from the deductive stack (§9.3) for three reasons:

Three reasons for quarantining Conjecture 8.3.2

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(1) It is speculative — requires proving quantum-causally
    disconnected sectors still gravitate. The framework has no
    current derivation of cross-sector burden coupling; the
    FIREWALL guardrail (Layer A probability vs Layer C gravity)
    prohibits probability from sourcing curvature, and cross-
    sector gravity would require either crossing the FIREWALL
    or identifying a new algebraic channel not present in the
    current framework.
(2) It is NOT the primary dark-matter mechanism. Relational
    burden T^(rel) is. The cross-sector conjecture is a
    secondary candidate at best, and not required to match
    observation if relational burden proves quantitatively
    sufficient (which it is expected to do, pending Q-1
    numerical simulation).
(3) It is explicitly quarantined from the deductive stack
    (§9.3) per Spec Ch. 12.4 anti-pattern (1): do not "tidy
    up" the quarantine list. The cross-sector conjecture is
    preserved verbatim from the Gen 3 Section_8 _2/_3 draft;
    un-quarantining early re-introduces the smuggling problem.
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FIREWALL boundary — cross-sector gravity crosses the FIREWALL

> The cross-sector gravity conjecture (§8.3.2) is the one place in Section 8 where the FIREWALL guardrail becomes operationally active. Branch weights p_k are probabilistic (Layer A); gravity is algebraic (Layer C). The conjecture that quantum-causally disconnected sectors gravitate through their burden tensors would require either (a) a probabilistic average over sectors sourcing curvature — which the FIREWALL prohibits — or (b) an algebraic cross-sector burden coupling that bypasses the probability layer entirely — which would require a new primitive not currently in the framework. **The conjecture is quarantined until one of these derivations closes.** No smuggling is permitted; the quarantine is preserved verbatim per Spec Ch. 12.4.

§8.3.3 Dark Energy as Expansive Burden

Layer C (conjectural)

Source: Section_8_Cosmology_2.txt §8.3.3 (Conjecture 8.3.3).

The observed accelerating expansion (dark energy) is the expansive pressure of open extension — the active SOE incorporation at the causal frontier generating new correlation volume. This is NOT a cosmological constant: $\Lambda_B = 0$ is exact (Theorem 5.5, restated in cosmological context as Theorem 8.5.1). It is the dynamic pressure of the universe continuously extending its own constraint-compatible volume at the SOE rate.

Conjecture 8.3.3 (Dark Energy = SOE Frontier Pressure).

The observed accelerating expansion is the expansive pressure of open extension — active SOE incorporation at the causal frontier generating new correlation volume. The effective dark-energy density $\rho_\Lambda^{\text{eff}}$ measured by ΛCDM fits is the time-averaged residual burden pressure:

$$\rho_\Lambda^{eff}(t) = \langle \rho_B^{residual}(t) \rangle \to 0 \text{ as } t \to \infty \tag{8.5}$$

As the universe expands and the causal horizon grows, new reconciliation capacity becomes available, but in the infinite-time limit, as $\mathbf{R}_t$ fully reconciles the universe, the residual pressure → 0. The universe asymptotically approaches $\Lambda = 0$. This is the structural resolution of the cosmological constant problem: there is no Λ to fine-tune, only a dynamic residual pressure that decays as reconciliation completes.

> **Q-2 (Dark energy = SOE frontier pressure) — PARTIAL**

> Quarantine item Q-2 is partially addressed by §8.3.3: **(i)** the mechanism — dark energy = SOE frontier pressure — is structurally derived from the SOE/MOE decomposition (P2 + P3); **(ii)** the dependency on $\Lambda_B = 0$ is CLOSED (Theorem 5.5, §5.5; restated as Theorem 8.5.1 in §8.5); **(iii)** the dependency on FLRW closure is PARTIAL — the Friedmann-like equation (8.4) is the Euler-Lagrange of MOE descent on FLRW (Conjecture 8.2.2), but the proof that MOE descent on the homogeneous isotropic metric yields exactly the Friedmann form is a structural conjecture, not yet a closed theorem. **What remains open**: quantitative recovery of the effective equation-of-state parameter $w \approx -1$ (consistent with ΛCDM observation) requires numerical simulation of MOE descent. Q-2 status: PARTIAL — Λ closed, FLRW partial, amplitudes open → §9 audit.

§8.4 Early Universe — Kinematic State, Inflation, Sector Formation

Layers A→C · P5

Source: Section_8_Cosmology base §8.4 + Section_8_Cosmology_2.txt §8.4. Resolves forward reference from §3.6.3; partially resolves forward reference from §7.4 (Q-4).

The early universe poses three questions: what was the initial state, what drove inflation, and how did reconciliation sectors form? RCF answers each structurally. The universe began in a pure kinematic state — the maximally coherent superposition across all constraint-satisfaction configurations, with maximal burden B_Δ. Cosmic inflation was the initial rapid phase of SOE extension at γ_{max}, with MOE descent exponentially suppressing cross-sub-sector coherence. Reconciliation sectors formed at different epochs as MOE descent crossed decoherence thresholds (Theorem 7.1). The three conjectures are tightly linked: the kinematic state supplies the initial condition, inflation supplies the rapid expansion, and sector formation supplies the decoherence structure that grounds the Born rule (§7.5) and the matter-antimatter asymmetry (Q-4).

§8.4.1 Initial Kinematic State

Layer A→B

Source: Section_8_Cosmology base §8.4.1 + Section_8_Cosmology_2.txt §8.4.1 (Conjecture 8.4.1). Resolves forward reference from §3.6.3.

Conjecture 8.4.1 (Initial Kinematic State).

The universe began in a pure kinematic state ω_{kin} on the full algebra $\mathscr{A}$ — a highly coherent superposition across all possible constraint-satisfaction configurations, with maximal B_Δ. This was the "big bang" as a reconciliation event: not a singularity (singularity avoidance is structural in RCF via the ℓ_0-floor, Theorem 5.4 / Q-14), but the initial state of maximum unreconciled burden from which MOE descent subsequently drives the universe toward the physical sector.

The kinematic state ω_{kin} is explicitly NOT pre-constrained to satisfy the master constraint (§0.2); the physical state emerges dynamically as MOE descent drives the system toward $\ker(\hat{M})$. At $t = 0$, the universe is "all potential, no actuality" — every constraint-satisfaction configuration is equally present in the superposition, and the burden $B_\Delta[\omega_{kin}]$ is maximal. The subsequent history of the universe is the history of MOE descent reducing B_Δ toward zero (the Master-Zero condition, Theorem 5.5).

Forward reference from §3.6.3 — RESOLVED

Section 3 v1.0 left a forward reference from §3.6.3 (cosmological initial condition → §8). **This merged Section 8 resolves that forward reference**: in §8.4.1 (P5), the initial kinematic state ω_{kin} is identified as the highly coherent superposition across constraint-satisfaction configurations with maximal B_Δ. The "big bang" is reconceived as a reconciliation event — the initial state of maximum unreconciled burden from which MOE descent drives the universe toward the physical sector. This is consistent with §3.6 (burden-clock potential Φ_C as the driver of cosmological time): the arrow of cosmological time points from ω_{kin} (maximal B_Δ) toward the asymptotic Master-Zero state. No new primitives are introduced; the initial state is the kinematic state of §0.2 evaluated on the full algebra.

§8.4.2 Inflation = Rapid SOE Extension at γ_{max} + MOE Dephasing

Layer A→C (conjectural)

Source: Section_8_Cosmology base §8.4.2 + Section_8_Cosmology_2.txt §8.4.2 (Conjecture 8.4.2).

Conjecture 8.4.2 (Inflation = Rapid SOE Extension).

Cosmic inflation was the initial phase of maximal SOE extension at rate γ_{max}, with MOE descent exponentially suppressing cross-sub-sector coherence. The rapid expansion is SOE frontier advance at the maximum SOE flux capacity (Corollary 8.1.2); the smoothing is MOE descent + dephasing (Theorem 7.1). The exponential expansion of inflation is the exponential suppression of cross-sector coherence — as MOE descent drives the kinematic superposition toward the physical sector, cross-sector burden $B_{cross}(i, j) \gg B_{intra}$ forces off-diagonal correlations to vanish (Thm 7.1), and the apparent super-luminal expansion is the SOE frontier rate γ_{max} operating on a rapidly fragmenting reconciliation landscape.

This conjecture resolves two puzzles of standard inflationary cosmology within the RCF framework: **(i)** the horizon problem (why is the CMB temperature uniform across causally disconnected regions?) — the answer is that the regions were NOT causally disconnected during inflation; they shared the same SOE frontier and the same kinematic superposition before MOE descent split them into separate sectors; **(ii)** the flatness problem (why is the spatial curvature k so close to zero?) — the answer is that rapid SOE extension at γ_{max} drives the reconciled volume $V(t) \propto a(t)^3$ to enormous values, diluting any initial curvature k/a^2 to negligible amplitude. Both resolutions are structural consequences of the SOE/MOE decomposition, not fine-tuned initial conditions.

Q-3 (Inflation = early-SOE rapid expansion) — OPEN

> Quarantine item Q-3 is structurally addressed in §8.4.2 but remains OPEN: **(i)** the mechanism — inflation = rapid SOE extension at γ_{max} + MOE dephasing — is structurally derived from the SOE/MOE decomposition (P2 + P5); **(ii)** the dependency on cosmological initial conditions is partially addressed by §8.4.1 (Conjecture 8.4.1, Initial Kinematic State), but the formal derivation of the inflationary epoch's duration, exit, and reheating mechanism remains separate from the core framework. **What remains open**: quantitative recovery of inflationary observables (scalar spectral index n_s, tensor-to-scalar ratio r, non-Gaussianity f_{NL}) requires a model of the SOE frontier fluctuation spectrum, which is not yet derived. Q-3 status: OPEN — structural mechanism proposed, quantitative recovery pending → §9 audit. Per Spec Ch. 12.2, Q-3 is "separate from core framework" — the inflationary epoch is a boundary condition on the cosmological initial state, not a theorem-target of the relational algebra.

§8.4.3 Sector Formation Epochs

Layer A→B

Source: Section_8_Cosmology base §8.4.3 + Section_8_Cosmology_2.txt §8.4.3 (Conjecture 8.4.3). Addresses forward reference from §7.4 (Q-4).

Conjecture 8.4.3 (Sector Formation Epochs).

Different reconciliation sectors formed at different epochs, corresponding to different constraint-obstruction thresholds being crossed as $\mathbf{R}_t$ operated. Our sector is one among many. In the SOE/MOE language, record sub-sectors formed as MOE descent crossed decoherence thresholds (Theorem 7.1): as the universe expanded and the burden B_Δ redistributed, cross-sector burden $B_{cross}(i, j)$ grew relative to intra-sector burden B_{intra}, triggering stable record separation (Thm 7.1) at specific epochs. The sector-formation timeline is therefore a cosmological observable, decodable from the decoherence structure of the present-day record landscape.

This conjecture connects §7.1 (stable record separation) to the cosmological timeline. The Born rule (Thm 7.5) operates over the record sectors that formed during these epochs; the sector weights $p_k = \mathrm{Tr}(\mathbb{P}_k\, \rho_{kin})$ are fixed by the kinematic state (§8.4.1) and revealed as MOE descent separates the sectors. The matter-antimatter asymmetry (Q-4) is a consequence of sector weight asymmetry: if the kinematic state ω_{kin} assigns slightly different weights p_k to matter-dominated and antimatter-dominated sectors, the observable asymmetry is fixed at the epoch of sector formation and preserved by the subsequent MOE descent.

> **Forward reference from §7.4 — PARTIALLY RESOLVED (Q-4)**

Section 7 v1.0 left a forward reference from §7.4 (sector weight asymmetry p_k → §8, Q-4 matter-antimatter asymmetry). **This merged Section 8 partially resolves that forward reference**: in §8.4.3 (P5), sector formation epochs are identified as the cosmological mechanism that fixes sector weights p_k — the kinematic state ω_{kin} assigns weights to potential sectors, and MOE descent crossing decoherence thresholds (Thm 7.1) reveals those weights as the sectors form. The Born rule (Thm 7.5, T-4 STRENGTHENED in §7) then operates over the formed sectors. **What remains open**: a formal derivation of the matter-antimatter weight asymmetry $\delta p = p_{matter} - p_{antimatter}$ from RCF primitives — the framework predicts that the asymmetry is fixed at sector formation, but does not yet derive its amplitude. Q-4 status: PARTIAL — T-4 closed (§7), mechanism proposed (§8.4.3), formal derivation pending → §9 audit.

§8.5 Cosmological Constant Revisited

Layer C · P6

Source: Section_8_Cosmology base §8.5 (Theorem 8.5.1) + Section 5 Theorem 5.5 ($\Lambda_B = 0$ EXACT).

The burden-derived cosmological constant is exactly zero. This was established in §5.5 as Theorem 5.5 (Λ_B = 0 EXACT), derived from the Master-Zero condition $\omega(\mathcal{M}) = 0$ being an exact asymptotic constraint achieved by MOE descent, not a tuned parameter. Section 8 restates this result in the cosmological context, drawing out its implications for the dark-energy problem and the asymptotic state of the universe.

Theorem 8.5.1 (Λ_B = 0 is Exact).

Conditional on T-2 (stable-mode assumption). Restates Theorem 5.5 in cosmological context.

The burden-derived cosmological constant is exactly zero (Theorem 5.3.1 / 5.5). The observed cosmic acceleration is not Λ but the time-dependent residual burden pressure $\rho_B^{residual}(t)$ (Conjecture 8.3.3). In the infinite-time limit, as $\mathbf{R}_t$ fully reconciles the universe, this pressure → 0:

$$\lim_{t\to\infty} \rho_B^{residual}(t) = 0 \Rightarrow \lim_{t\to\infty} \Lambda_{eff}(t) = 0 \tag{8.6}$$

The universe asymptotically approaches Λ = 0. This is the structural resolution of the cosmological constant problem: there is no Λ to fine-tune, because the Master-Zero condition $\omega(\mathcal{M}) = 0$ is an EXACT asymptotic constraint achieved by MOE descent (§5.5, Theorem 5.5), not a tuned parameter. The 120-order-of-magnitude discrepancy between the naive QFT vacuum energy estimate and the observed dark-energy density does not arise in RCF, because RCF does not quantize fields on a fixed background — fields emerge as burden-flux excitations (§4.3.5), and the burden vacuum is exactly zero by the Master-Zero condition.

Q-9 (Cosmological constant problem) — PARTIAL

> Quarantine item Q-9 (the 120-order-of-magnitude cosmological constant problem) is structurally addressed by §8.5 / §5.5: **(i)** $\Lambda_B = 0$ is EXACT (Theorem 5.5, restated as Theorem 8.5.1) — there is no Λ to fine-tune, because the Master-Zero condition ω(M) = 0 is an exact asymptotic constraint achieved by MOE descent; **(ii)** the apparent dark-energy density is reinterpreted as dynamic residual burden pressure (Conjecture 8.3.3), not a vacuum energy; **(iii)** the asymptotic limit $\Lambda_{eff} \to 0$ as $t \to \infty$ is structurally forced. **What remains open**: the formal closure of Q-9 requires T-2 (stable-mode assumption) — the proof that the Master-Zero condition is achieved asymptotically by MOE descent under the stable-mode assumption. T-2 is currently a theorem-target, not a closed theorem. Q-9 status: PARTIAL — Λ_B=0 closed (conditional on T-2), T-2 pending → §9 audit.

The cosmological constant problem is, in RCF terms, a category error of the standard frameworks: it asks "what is the value of Λ?" when the correct question is "what is the residual burden pressure, and how does it decay as reconciliation completes?" The answer to the latter is structural — the residual pressure is the SOE frontier pressure (Conjecture 8.3.3), and it decays to zero as the universe asymptotically approaches full reconciliation.

§8.6 ΛCDM Recovery, Lorentz IR, Q-4 Status, and Nine Guardrails

Layers A→C · P7

Source: Status tables DERIVED from §8.1–§8.5 + nine guardrails PORT from Section_8_2 architectural-position table.

Section 8 closes with three status tables summarizing the partial recovery of standard cosmology (Q-13), the status of Lorentz invariance in the deep IR (Q-17), and the status of the matter-antimatter asymmetry (Q-4). Each table records what is structurally closed, what is partially closed, and what remains open — with explicit dependency contracts for the open items. The nine consolidated guardrails codify the architectural discipline of the cosmology layer.

§8.6.1 Q-13: ΛCDM Recovery Status

ΛCDM Component	RCF Mechanism	Source	Status
FLRW metric	MOE descent on FLRW (Conj 8.2.2)	§8.2 (P3)	STRUCTURALLY CLOSED — Friedmann form derived as Euler-Lagrange
Spatial curvature k	Sector geometry (Conj 8.2.2)	§8.2 (P3)	OPEN — k determined by initial kinematic state (Q-3)
Matter density Ω_m	Mode burden $\rho_B^{(mode)}$	§5.1.2 + §8.2	STRUCTURALLY CLOSED — mode channel contributes to ρ_B
Dark matter Ω_{DM}	Relational burden $\rho_B^{(rel)}$	§8.3.1 (P4)	PARTIAL — structural match; amplitudes pending Q-1

ΛCDM Component	RCF Mechanism	Source	Status
Dark energy Ω_Λ	SOE frontier pressure (NOT Λ; Λ_B=0 exact)	§8.3.3 + §8.5	PARTIAL — Λ closed, effective equation-of-state w ≈ −1 pending
Cosmological constant Λ	Λ_B = 0 EXACT (Thm 5.5 / 8.5.1)	§5.5 + §8.5 (P6)	CLOSED (conditional on T-2) — framework predicts Λ = 0
Inflationary epoch	Rapid SOE extension at γ_{max} + MOE dephasing	§8.4.2 (P5)	OPEN — Q-3; structural mechanism only, amplitudes pending
H_0 tension	SOE frontier rate coarse-grained	§8.1 (P2)	OPEN — requires numerical simulation of MOE descent on FLRW

Table 8.6.1 — Q-13 ΛCDM recovery status. The framework recovers the ΛCDM FORM (FLRW closure, dark sector structural mechanisms, Λ = 0 prediction), but not yet the AMPLITUDES (H_0, Ω_m, Ω_{DM}, Ω_Λ^{eff}) — those require numerical simulation of MOE descent on FLRW plus closure of Q-1, Q-2, Q-3.

§8.6.2 Q-17: Lorentz Invariance in Deep IR Status

Component	RCF Mechanism	Source	Status
Universal speed limit c	Two-scale speed limit $c = \gamma \cdot \ell_0$ (T-1)	§1.3	CLOSED — derived as SOE/MOE speed ceiling
Lorentzian signature (−,+,+,+)	Positive spatial correlation + negative burden lapse (Thm 5.3)	§5.3	CLOSED — derived as emergent metric signature
Lorentz symmetry (full group)	Deep-IR limit of correlation geometry	§8.1 (P2) + §5.3	PARTIAL — signature closed, full group equivalence pending
Equivalence to special relativity	Deep-IR limit of emergent metric tensor	§8.1 + §5.3	OPEN — formal proof that correlation geometry → Minkowski

Table 8.6.2 — Q-17 Lorentz invariance in deep IR status. The speed limit and signature are structurally closed; full Lorentz group equivalence and SR identification remain open as theorem-targets.

§8.6.3 Q-4: Matter-Antimatter Asymmetry Status

Component	RCF Mechanism	Source	Status
Born rule (T-4)	Z-envariance as MOE fixed-point symmetry	§7.5 (Thm 7.5)	CLOSED — T-4 STRENGTHENED
Sector weights p_k	$\mathrm{Tr}(P_k\, \rho_{kin})$ from kinematic state	§7.4 + §8.4.1	STRUCTURALLY CLOSED — formal probability layer complete
Sector formation mechanism	MOE descent crossing decoherence thresholds (Thm 7.1)	§8.4.3 (P5)	PARTIAL — mechanism proposed, formal derivation pending
Asymmetry amplitude δp	Kinematic state weight difference matter vs antimatter	§8.4.1 + §8.4.3	OPEN — framework predicts asymmetry fixed at sector formation; amplitude derivation pending

Table 8.6.3 — Q-4 matter-antimatter asymmetry status. The Born rule (T-4) and sector weight structure are closed in §7; sector formation mechanism proposed in §8.4.3; formal derivation of the asymmetry amplitude remains open as a theorem-target.

§8.6.4 Nine Consolidated Guardrails for the Cosmology Layer

The cosmology layer is governed by nine consolidated guardrails. These codify the architectural discipline established across Sections 0–7 and reaffirmed in Section 8:

Nine Consolidated Guardrails for the Cosmology Layer

```
(1) Cosmological scale is Layer B→C (SOE/MOE), not Layer A.
    Probability (Layer A) does not source cosmic curvature
    (Layer C). The FIREWALL guardrail (§5.0/§6.0/§7.0 P1) is
    REAFFIRMED for cosmology.

(2) FIREWALL — branch weights p_k are PROBABILISTIC (Layer A),
    burden is ALGEBRAIC (Layer A/B/C). Burden linearity
    (§0.3 Property 3) is a PROVEN IDENTITY of the linear
    functional, NOT an averaging-over-outcomes applied to
    source curvature. Tr(ρ_kin F̂) = Σ_k p_k Tr(ρ_k F̂).

(3) Λ_B = 0 EXACT (Theorem 5.5, restated as Theorem 8.5.1).
    No cosmological constant problem — there is no Λ to
    fine-tune. The 120-order-of-magnitude discrepancy of
    standard QFT does not arise because RCF does not quantize
    fields on a fixed background.

(4) Dark matter is RELATIONAL BURDEN (derived, T^(rel) =
    [Ĉ_α, Π̂_net]). Cross-sector gravity (§8.3.2) is
    QUARANTINED — speculative, NOT primary mechanism, isolated
    from deductive stack (§9.3).

(5) Dark energy is SOE FRONTIER PRESSURE (Conjecture 8.3.3).
    NOT a cosmological constant. Dynamic residual burden
    pressure that decays to zero as reconciliation completes.

(6) Inflation is RAPID SOE EXTENSION at γ_max + MOE dephasing
```

```
    (Conjecture 8.4.2). Quarantine Q-3: depends on
    cosmological initial conditions; separate from core
    framework.

(7) Initial kinematic state ω_kin is STRUCTURAL CONJECTURE
    (Conjecture 8.4.1). Resolves §3.6.3 forward reference.
    Maximal B_Δ; universe begins in maximum unreconciled
    burden. NOT a singularity — singularity avoidance is
    structural via ℓ_0-floor (Thm 5.4 / Q-14).

(8) Quarantine discipline PRESERVED VERBATIM from Spec Ch.
    12.2. The 17-point list is the one part of Gen 3 carried
    over unchanged. Do NOT "tidy up" the quarantine list
    (Spec Ch. 12.4 anti-pattern 1). Un-quarantining early
    re-introduces the smuggling problem.

(9) NO SMUGGLING — every claim about physics beyond the
    framework's core (dark sector, Standard Model, quantum
    gravity, inflationary observables) is QUARANTINED with a
    clear closure path. The canonical manuscript, when
    complete, will be the first version of RCF in which every
    definition sits at the layer where its ingredients first
    exist, every theorem either has a proof or is explicitly
    flagged as a target, and every claim about physics beyond
    the core is quarantined.
```

The forward-reference contract is complete: four forward references from Sections 2, 3, 5, and 7 are RESOLVED here (§2.10 cubic correlation volume, §3.6.3 cosmological initial condition, §5.1.2 dark matter = relational burden, §5.4.2 $\Lambda_B = 0$ + dark energy); one is PARTIALLY resolved (§7.4 sector weight asymmetry, Q-4). Seven cosmology-relevant quarantine items (Q-1, Q-2, Q-3, Q-4, Q-9, Q-13, Q-17) are surfaced with explicit dependency tags. All forward references are one-way; there is no circularity. Section 8 does not modify any structure of Sections 0–7.

§8.7 Architectural Summary

The following table summarizes the structural units established in Section 8, with their layer, source, and status. The table confirms the acyclic emergence chain: open extension (§1.4) → cosmic expansion = SOE rate → cubic volume growth → Friedmann = MOE descent on FLRW → dark sector (relational burden + quarantined cross-sector + dark energy) → early universe (kinematic state + inflation + sector formation) → $\Lambda_B = 0$ EXACT → ΛCDM/Lorentz IR/Q-4 status.

#	Structural Unit	Layer	Source	Status / Notes
8.0	Layer placement + FIREWALL (P1)	B→C, A quarantined	Sec_8_2 §8.0 + Spec 12.2	P1 — seven Q-items surfaced; FIREWALL reaffirmed
8.1	Expansion = SOE Frontier Rate (Conj 8.1.1)	B→C	Sec_8_2 §8.1	P2 — $H(t) = \gamma_{eff}(t)$; RESOLVES §2.10 forward ref
8.2	Expansion Rate Limit (Cor 8.1.2)	B→C	Sec_8_2 §8.1	P2 — $H(t) \leq \gamma_{max} = 1/\varepsilon$; SOE flux capacity bound
8.3	Cubic Volume Growth (Conj 8.2.1)	B→C	Sec_8_2 §8.2	P3 — $V(t) \propto a(t)^3$ from SOE-invariant D=3 (Thm 2.3.3)
8.4	Friedmann from MOE Descent (Conj 8.2.2)	C	Sec_8_2 §8.2	P3 — Euler-Lagrange of MOE on FLRW; $\Lambda_B = 0$ forces no-Λ form; RESOLVES §5.4.2 forward ref
8.5	Relational Burden = Dark Matter (§8.3.1)	B→C (derived)	Sec_8_1 §8.3.1 + Sec_8_2 §8.3.1	P4 — $T^{(rel)} = [\hat{C}_\alpha, \Pi_{net}]$; 4 properties; RESOLVES §5.1.2 forward ref; Q-1 OPEN
8.6	Cross-Sector Gravity (Conj 8.3.2)	A (speculative)	Sec_8_1 §8.3.2 + Sec_8_2 §8.3.2	P4 — QUARANTINED; crosses FIREWALL; isolated from deductive stack (§9.3)
8.7	Dark Energy = SOE Frontier Pressure (Conj 8.3.3)	C	Sec_8_2 §8.3.3	P4 — NOT cosmological constant; $\Lambda_B = 0$ exact; Q-2 PARTIAL
8.8	Initial Kinematic State (Conj 8.4.1)	A→B	Sec_8 base §8.4.1 + Sec_8_2 §8.4.1	P5 — ω_{kin} maximal B_Δ; RESOLVES §3.6.3 forward ref
8.9	Inflation = Rapid SOE Extension (Conj 8.4.2)	A→C	Sec_8 base §8.4.2 + Sec_8_2 §8.4.2	P5 — γ_{max} + MOE dephasing; Q-3 OPEN (separate from core)
8.10	Sector Formation Epochs (Conj 8.4.3)	A→B	Sec_8 base §8.4.3 + Sec_8_2 §8.4.3	P5 — MOE descent crosses decoherence thresholds; PARTIALLY RESOLVES §7.4 (Q-4)
8.11	$\Lambda_B = 0$ EXACT (Thm 8.5.1)	C	Sec_8 base §8.5 + §5.5 (Thm 5.5)	P6 — restates Thm 5.5 cosmologically; Q-9 PARTIAL (T-2 pending)

#	Structural Unit	Layer	Source	Status / Notes
8.12	ΛCDM Recovery Status (Table 8.6.1)	B→C	DERIVED §8.1–§8.5	P7 — Q-13 PARTIAL: FLRW closed, dark sector amplitudes open
8.13	Lorentz IR Status (Table 8.6.2)	B→C	DERIVED §8.1 + §5.3	P7 — Q-17 PARTIAL: c and signature closed, full Lorentz equivalence pending
8.14	Q-4 Asymmetry Status (Table 8.6.3)	A→C	DERIVED §7 + §8.4	P7 — Q-4 PARTIAL: T-4 closed, mechanism proposed, amplitude pending
8.15	Nine Consolidated Guardrails	A→C	PORT Sec_8_2 + Spec 12.4	P7 — FIREWALL, Λ_B = 0, quarantine discipline, no-smuggling

Table 8.7.1 — Fifteen structural units of Section 8, by layer, source, and status.

The cumulative architectural chain of Section 8 is:

Conceptual chain of Section 8

```
open extension (§1.4) at causal frontier
        |
        ▼
cosmic expansion H(t) = γ_eff(t) = (1/V)·dV/dt  (Conj 8.1.1, P2)
        |   bounded by H(t) ≤ γ_max = 1/ε  (Cor 8.1.2)
        |   RESOLVES §2.10 forward ref (cubic □_ω ↔ a(t))
        ▼
cubic volume growth V(t) ∝ a(t)³  (Conj 8.2.1, P3)
        |   from SOE-invariant D=3 (Thm 2.3.3)
        ▼
Friedmann (ȧ/a)² = (8πG/3)·ρ_B − k/a²  (Conj 8.2.2, P3)
        |   Euler-Lagrange of MOE descent on FLRW
        |   Λ_B = 0 forces NO-Λ form (Thm 5.5)
        |   RESOLVES §5.4.2 forward ref
        ▼
dark sector (three-part decomposition, P4):
   ├─ §8.3.1 Primary: T^(rel) = [Ĉ_α, Π̂_net]  (RESOLVES §5.1.2; Q-1 OPEN)
```

```
    ├─ §8.3.2 Secondary: cross-sector gravity  (QUARANTINED, crosses FIREWALL)
    └─ §8.3.3 Dark energy = SOE frontier pressure  (NOT Λ; Q-2 PARTIAL)
         │
         ▼
early universe (P5):
    ├─ §8.4.1 Initial kinematic state ω_kin  (RESOLVES §3.6.3)
    ├─ §8.4.2 Inflation = rapid SOE at γ_max + MOE dephasing  (Q-3 OPEN)
    └─ §8.4.3 Sector formation epochs  (PARTIALLY RESOLVES §7.4 / Q-4)
         │
         ▼
Λ_B = 0 EXACT  (Thm 8.5.1, P6; restates Thm 5.5)
         │   Q-9 PARTIAL (T-2 stable-mode pending)
         ▼
ΛCDM/Lorentz IR/Q-4 status tables + nine guardrails  (P7)
         │   Q-13 PARTIAL · Q-17 PARTIAL · Q-4 PARTIAL
         ▼
open quarantine items deferred to §9 audit
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Forward-reference resolution summary

Section 8 RESOLVES four forward references from prior sections: **(a)** §2.10 cubic correlation volume $\mathcal{V}_\omega$ ↔ cosmological scale factor a(t) — resolved in §8.1 (P2); **(b)** §3.6.3 cosmological initial condition — resolved in §8.4.1 (P5); **(c)** §5.1.2 dark matter = relational burden — resolved in §8.3.1 (P4); **(d)** §5.4.2 $\Lambda_B = 0$ + dark energy reinterpretation — resolved in §8.2 + §8.3.3 + §8.5 (P3, P4, P6). One forward reference is PARTIALLY resolved: **(e)** §7.4 sector weight asymmetry p_k (Q-4) — sector formation epochs (§8.4.3) supply the structural mechanism, but formal derivation of the asymmetry amplitude remains open. **Open quarantine items deferred to §9 audit**: Q-1 (dark matter amplitudes, T-7 + numerical sim), Q-2 (dark energy effective w, FLRW formal closure), Q-3 (inflationary observables, separate from core), Q-4 (asymmetry amplitude derivation), Q-9 (formal closure of $\Lambda_B = 0$ conditional on T-2), Q-13 (ΛCDM amplitudes), Q-17 (full Lorentz group equivalence).

This completes the cosmology layer. The framework now possesses an emergent spacetime, populated by interacting matter, governed by gravity, capable of explaining definite observational outcomes, protected by the FIREWALL guardrail, and applied at the largest cosmological scales — with cosmic expansion as SOE frontier rate, Friedmann dynamics as MOE descent on FLRW, dark matter as relational burden, dark energy as SOE frontier pressure (NOT Λ), and $\Lambda_B = 0$ EXACT structurally resolving the cosmological constant problem. The final step is audit: Section 9 will close the 17-point quarantine list with explicit closure paths, document the theorem-target dependency graph, and verify that the cumulative architecture (Sections 0–8) is acyclic, with every claim traced to a source, a closed theorem, or an explicitly labeled quarantine item.